ForgeSlicer

Comprehensive Manual QA Test Plan

Version 1.6 · June 27, 2026

Environment: **https://orca-cad-slice.preview.emergentagent.com**

ForgeSlicer is a browser-based 3D CAD + Slicer that fuses a TinkerCAD-style primitive editor, Shapr3D-style reverse engineering, and an OrcaSlicer-style slicing handoff into a single workflow. This document is the comprehensive manual QA test plan that walks a tester through every user-facing surface — primitives, Boolean operations, importers, RANSAC reverse engineering, AI / voice workflows, the community gallery, Learn lessons, Trust pages, account and email flows, and slicer handoff.

Each test case has a stable ID (so regressions can be referenced across releases), the click-by-click steps a tester should follow, the expected result, and a priority. Run P0 cases on every release, P1 cases weekly, and P2 cases when a related area changes.

Test environment: **https://orca-cad-slice.preview.emergentagent.com**. Recommended browsers: latest Chrome, Firefox, and Safari. Use Chromium-based browsers for WebGPU-dependent paths.

P0 Run every release	**P1** Run weekly	**P2** Run on related changes

Contents

1. Onboarding & First-Run

Landing tab bar (iter-108)

ID	Test Case	Steps	Expected Result	Pri
LAND-01	Landing tab bar shows 6 tabs in fixed order directly below the header	1. Open / in a private window. 2. Locate the row of tabs at the top of the page (right under the site header).	Exactly 6 tabs in this order: Home · Start · Templates · Gallery · Learn · Trust. Each has an icon + label and a data-testid 'landing-tab-{id}'. The container has data-testid 'landing-tabbar'. Site header is above, footer below.	P0
LAND-02	Home is the default tab on every fresh load	1. Open / in a private window. 2. Observe which tab is active before clicking anything.	Home tab is selected (orange underline + orange text); active panel data-testid is 'landing-tabpanel-home'. The SsoBridge banner, hero headline 'Design. Speak. Slice. Print.', anvil image, CTA buttons, and stats row all render.	P0
LAND-03	Switching tabs swaps the entire body content	1. Click each tab in turn: Home → Start → Templates → Gallery → Learn → Trust.	Each click swaps the tabpanel content within ~50 ms. When NOT on Home, the hero is NOT visible. When NOT on Trust, the trust tab content is NOT visible. No console errors during the switch.	P0
LAND-04	Tab state is session-only — refresh resets to Home	1. Click Trust tab. 2. Hard refresh the page (Cmd-R / F5).	After refresh, the URL stays '/', Home tab is selected again. No ?tab= query string is added or required.	P1
LAND-05	Trust tab links route to the full Trust hub, Privacy, and Changelog	1. Switch to Trust tab. 2. Click each of the 3 CTAs.	'Read the full Trust hub' → /trust. 'Privacy details' → /privacy. 'Changelog' → /changelog. Each destination renders correctly.	P1
LAND-06	Hero 'Try an Example Project' button switches to the Templates tab	1. On Home tab, locate the 'Try an Example Project' CTA in the hero. 2. Click it.	Templates tab becomes active (orange underline moves), the tabpanel swaps to show the 8-card 'Start with a finished part' grid, and the viewport scrolls to put the tab bar at the top. No console errors. Regression check: this used to be a no-op when the underlying scroll targets moved into other tabs.	P0

Beginner Starters

ID	Test Case	Steps	Expected Result	Pri
ONB-01	12 Beginner Starter cards render in a responsive grid inside the Start tab	1. Open / in a private window. 2. Ensure the Start tab is active (it is by default). 3. Scroll down past the marketing sections (AI/voice, Audience, Feature grid, 5-step) until you reach the 'Beginner Starter Projects' grid. 4. Count the cards.	12 starter cards render in a responsive grid (2 cols on small, 3 on lg, 4 on xl). Each card shows an icon, title, difficulty pill, estimated print time, and skill tags. data-testid 'landing-beginner-starters' is present inside 'landing-tabpanel-start' and 'landing-starters-grid' contains 12 children.	P0
ONB-02	Opening a starter populates the workspace with that starter's geometry	1. From the Beginner Starters grid, click any card (e.g. 'Keychain'). 2. Wait for /workspace?template=<id> to mount.	URL is /workspace?template=<id>; sessionStorage.forgeslicer.launchTemplate contains the payload; the starter mesh appears in the scene; the outliner shows the starter's nodes.	P0

ID	Test Case	Steps	Expected Result	Pri
ONB-03	Skip onboarding banner is dismissible & non-blocking	1. Land on /. 2. Locate the dismissible announcement banner in the bottom-right corner on product routes. 3. Click the X.	Banner closes, never reappears in the same session, no fullscreen modal interrupts work.	**P1**
ONB-04	Workspace tips toast cycles one tip at a time on entry	1. Sign in. 2. Navigate to /workspace from the landing CTA. 3. Wait for the small bottom-right toast.	A small dismissible 'tip of the day'-style toast appears (NOT a fullscreen popup) with the next unseen tip. Closing it advances the queue; reopening /workspace later shows the next tip. After all tips are seen, a 'That's all the tips for now.' toast fires once.	**P1**
ONB-05	After creating an account, the user lands on the landing page (not the workspace)	1. Go to /signin?mode=register. 2. Fill display name, email, password. 3. Click 'Create account'.	After success, URL becomes '/' and the landing page renders with the user signed in (avatar visible in header). It MUST NOT auto-redirect to /workspace. Same expectation for Google sign-in and magic-link completion when no explicit ?return= was specified.	**P0**

Auth (email + Google)

ID	Test Case	Steps	Expected Result	Pri
AUTH-01	Email + password signup → email verification	1. /signup. 2. Use a fresh address. 3. Submit. 4. Check inbox.	Account row created, verification email arrives via Resend, link sets `verified=true`.	**P0**
AUTH-02	Magic link sign-in	1. /signin. 2. Click 'Send magic link'. 3. Click link in email.	Single-use link signs the user in within 15 minutes; reusing it returns an expired error.	**P0**
AUTH-03	Password reset	1. /signin → 'Forgot password'. 2. Submit email. 3. Open link. 4. Set new password.	Reset link valid 60 min, signs user out of all sessions on success.	**P0**
AUTH-04	Emergent-managed Google sign-in	1. /signin → 'Continue with Google'. 2. Approve the consent screen.	User is created or matched on email, profile photo + name populate.	**P0**
AUTH-05	Google sign-in completes inside the 45 s frontend timeout (no spurious cancellation)	1. From / click any Templates card while signed out → land on /signin. 2. Click 'Continue with Google' and complete the Google consent screen. 3. Watch the 'Signing you in…' screen.	Within ≤ 35 seconds the backend exchange returns 200 (worst case is 4 × 7 s httpx + ~5.4 s backoff = 33.4 s — comfortably under the 45 s axios ceiling in `lib/auth.js -> authApi.exchange`). The user is redirected to the page that initiated sign-in. The browser network tab does NOT show `(canceled)` on the `POST /api/auth/session` row. Regression: the earlier 15 s per-attempt budget could push total backend time past 60 s, surfacing as 'timeout of 45000ms exceeded' on the client.	**P0**

ID	Test Case	Steps	Expected Result	Pri
AUTH-06	Backend logs per-attempt timing for auth-provider exchange	1. After a Google sign-in (success or failure). 2. Inspect backend.out.log / backend.err.log.	INFO lines of the form 'auth-provider attempt N returned <status> in <X.XX>s; retrying' for each retry, and either 'auth-provider succeeded on attempt N in X.XXs (total Y.YYs)' on success or 'auth-provider gave up after 4 attempts in Y.YYs (last_status=…)' on failure. These let an operator confirm whether the upstream provider or our own retry policy is the culprit if a user complains.	**P1**

2. Primitive Editor & Boolean Operations

Primitives

ID	Test Case	Steps	Expected Result	Pri
PRIM-01	Insert cube / sphere / cylinder / cone / torus / triangle	1. Editor → Primitives palette. 2. Click each shape in turn.	Each primitive spawns at origin with sensible defaults; scene tree updates; gizmo attaches.	**P0**
PRIM-02	Edit a primitive's dimensions via the right-side Inspector (numeric fields)	1. Select a cylinder by clicking it. 2. Open the right-side Inspector panel (RightPanel). 3. Edit Width / Depth / Height numeric inputs, then radius and segment count if exposed.	Mesh regenerates live as you type (debounced), without flicker or selection loss. Each commit creates one undo step. The dimensions reflect back into the gizmo's bounding box.	**P0**
PRIM-02b	Edit a primitive by dragging its TinkerCAD-style face handles directly	1. Select a cube. 2. Grab one of the small colored squares on a face (cyan / magenta / yellow / etc). 3. Drag outward and release.	The face moves along its normal in real time; opposite face stays put; final dimensions match the drag delta (±0.1 mm with snapping on). One undo step recorded for the whole drag.	**P0**
PRIM-03	Triangle primitive — equilateral default	1. Insert triangle. 2. Inspect dimensions panel.	Equilateral by default; PRD note: configurable base/height/angles is upcoming P1.	**P1**
PRIM-04	Transform: move / rotate / scale via gizmo	1. Select primitive. 2. Press G / R / S (or click toolbar). 3. Drag axes.	Snapping to 1 mm / 15° works with Shift; numeric panel reflects gizmo state.	**P0**

Boolean Operations (Manifold-3D WASM)

ID	Test Case	Steps	Expected Result	Pri
BOOL-01	Union two overlapping cubes	1. Insert 2 cubes overlapping. 2. Multi-select. 3. Toolbar → Union.	Result is a single watertight mesh; volume ≈ sum minus overlap; history step recorded.	**P0**
BOOL-02	Subtract cylinder from cube (hole)	1. Insert cube and cylinder passing through it. 2. Select cube then cylinder. 3. Subtract.	Hole punched cleanly; resulting mesh is manifold; no inverted normals.	**P0**
BOOL-03	Intersect cube ∩ sphere	1. Place a sphere overlapping a cube. 2. Intersect.	Yields the cube-corner-cut-by-sphere lens; mesh is closed; saves to project.	**P0**
BOOL-04	Boolean on non-manifold input shows healing toast	1. Import a known non-manifold STL. 2. Attempt Boolean.	Healing pass runs; if unrecoverable, user sees clear error toast instead of a crash.	**P1**

3. Importers (STL / OBJ / 3MF / SVG / ZIP)

Mesh imports

ID	Test Case	Steps	Expected Result	Pri
IMP-01	STL drag-drop	1. Drag any STL onto the canvas.	Mesh appears centered, axis-aligned bbox shown; units default mm.	**P0**
IMP-02	OBJ with material groups	1. Import an OBJ that has multiple groups.	Groups become separate scene-tree nodes preserving names.	**P0**
IMP-03	3MF multi-object project	1. Import a 3MF authored in PrusaSlicer or Bambu Studio.	All objects retained with their original transforms; print-settings warning if present.	**P0**
IMP-04	SVG extrude	1. Drop an SVG. 2. Set extrude depth in the modal.	2D paths become a solid mesh; bezier curves sampled smoothly.	**P1**
IMP-05	ZIP archive (mixed assets)	1. Drop a ZIP containing STL + 3MF + SVG.	Importer enumerates contents, asks the user to choose, imports the selection.	**P1**

4. Reverse Engineering (RANSAC)

Dialog & detection (Phase 1–3 shipped)

ID	Test Case	Steps	Expected Result	Pri
RE-01	Open Reverse Engineer dialog from a selected mesh	1. Import an STL (e.g. MiniRack_RPI4_Tray). 2. Select the imported mesh in the viewport or Outliner. 3. Toolbar → 'Reverse engineer'. NOTE: if you see 'Segmentation failed (HTTP 404)', do a hard refresh (Ctrl-Shift-R / Cmd-Shift-R) — that error means your browser cached an older frontend bundle from before the /api/mesh/segment route shipped; the server is fine.	Modal opens with title 'Reverse Engineer', subtitle 'RANSAC primitive detection on <mesh name>', a stats row (Planes / Cylinders / Spheres / Coverage), and a list of detected primitives. data-testid 'reverse-engineer-dialog' is present.	**P0**
RE-02	RANSAC detects a box+cylinder composite	1. Import the 'bracket.stl' sample. 2. Run detection.	Result lists at least 1 box and 1 cylinder with sane dimensions (±5%).	**P0**
RE-03	Sensitivity slider (Phase 4, upcoming)	1. Drag the sensitivity slider. 2. Re-run detection.	Higher sensitivity yields more primitives, lower yields fewer; no infinite spinner.	**P1**
RE-04	Replace with primitives button (Phase 5, upcoming)	1. Click 'Replace with primitives'.	Original mesh hidden, parametric Three.js primitives spawned at detected transforms; undo restores mesh.	**P0**

5. AI & Voice Workflows

Conversational AI

ID	Test Case	Steps	Expected Result	Pri
AI-01	Voice command: 'make a 20 mm cube'	1. Click mic. 2. Speak the command.	Whisper transcribes; GPT-5.2 returns JSON intent; cube of side 20 mm appears.	**P0**
AI-02	Meshy AI: text → 3D	1. AI panel → 'Text to 3D'. 2. 'low-poly rocket'. 3. Submit.	Job polls Meshy; result mesh imports into the scene when ready; failure surfaces friendly toast.	**P0**
AI-03	Meshy AI: image → 3D	1. Upload an image to the AI panel. 2. Generate.	Returns watertight mesh; preview thumbnail accurate.	**P1**
AI-04	Meshy attribution copy	1. Open AI panel.	Footer explicitly states 'Powered by Meshy.ai — third-party service'.	**P1**

Text on Surface

ID	Test Case	Steps	Expected Result	Pri
TXT-01	Text-on-surface MVP	1. Select a flat face. 2. Tools → Text on surface. 3. Type 'Hi'. 4. Apply.	Helvetiker text mesh extrudes onto the face; aligned to face normal; Boolean-union onto host.	**P0**

6. Measurement, Snapping & Grid

Measurement

ID	Test Case	Steps	Expected Result	Pri
MEAS-01	Linear measurement between two vertices	1. Tool → Measure. 2. Click 2 vertices.	Distance label shown in mm; persists until cleared; respects unit toggle.	**P1**
MEAS-02	Grid + snapping toggles	1. View menu → Toggle grid / Toggle snapping.	Grid renders at 10 mm; snapping locks transforms to 1 mm; toggles persist across reloads.	**P1**

7. Community Gallery (v2)

Browse & filter

ID	Test Case	Steps	Expected Result	Pri
GAL-01	Public gallery loads with taxonomy	1. Open /gallery.	Categories sidebar visible (Functional, Decorative, Educational…), pagination works, featured creators row populated.	**P0**
GAL-02	Filter by tag + category	1. Select category 'Functional' + tag 'organizer'.	Result set narrows; URL reflects state; deep link reproducible.	**P0**
GAL-03	Visibility: private item hidden in public gallery	1. As user A publish private item. 2. Sign out. 3. Browse gallery.	Private item NOT listed publicly but still appears in A's profile.	**P0**
GAL-04	Contributor Lifetime celebration email	1. As an account, cross 100 components + 20 designs.	Resend dispatches the celebration email; badge appears on profile.	**P1**

8. Learn Section

Lessons

ID	Test Case	Steps	Expected Result	Pri
LRN-01	/learn lists 8 lessons	1. Open /learn.	8 lesson cards render with duration + difficulty tags.	**P1**
LRN-02	Lesson detail page deep-links	1. Click a lesson. 2. Copy URL. 3. Open in incognito.	Lesson loads directly, meta tags + structured data correct.	**P1**

9. Trust, Privacy & Changelog

Trust pages

ID	Test Case	Steps	Expected Result	Pri
TRUST-01	/trust renders policy summary	1. Open /trust.	Sections: data handling, third-party services (Meshy, OpenAI, Resend), security posture.	**P1**
TRUST-02	/privacy + /changelog accessible from footer	1. Scroll to footer on any page.	Both links present and resolve to readable pages.	**P1**

10. Slicer Handoff

Multi-object 3MF export → OrcaSlicer

ID	Test Case	Steps	Expected Result	Pri
SLI-01	Send to OrcaSlicer (desktop)	1. Build a scene with 3 objects. 2. Toolbar → 'Send to slicer' → Orca.	3MF downloads with all objects, transforms, and modifier volumes; opens in Orca without warnings.	**P0**
SLI-02	PrusaSlicer / Bambu Studio variants	1. Repeat with Prusa and Bambu targets.	Each slicer opens the 3MF cleanly; vendor-specific profile metadata respected when set.	**P1**
SLI-03	Watertight check before export	1. Force a non-manifold mesh. 2. Attempt export.	Modal warns and offers auto-heal; export blocked until user accepts or cancels.	**P0**

11. SEO & Marketing Surfaces

SEO landings

ID	Test Case	Steps	Expected Result	Pri
SEO-01	Targeted landings render unique meta	1. Visit each of the 8 SEO pages from /seo (or sitemap).	Each page has unique <title>, <meta description>, OG tags, JSON-LD.	**P2**
SEO-02	sitemap.xml + robots.txt valid	1. curl /sitemap.xml and /robots.txt.	XML validates, robots references the sitemap, no 5xx.	**P2**

12. Cross-Cutting Quality Bars

Performance & accessibility

ID	Test Case	Steps	Expected Result	Pri
QA-01	Cold load TTI < 4 s on cable	1. Lighthouse on /editor.	Performance score ≥ 80, no render-blocking JS warnings.	**P1**
QA-02	Keyboard navigation across modals	1. Tab through Reverse Engineer dialog.	Focus trap honored, ESC closes, no focus-loss bugs.	**P1**
QA-03	Mobile responsive sanity	1. Open / on a 390 px-wide viewport.	Hero, gallery, learn render without horizontal scroll; editor shows a gentle 'desktop recommended' note.	**P2**

Appendix A — Lexicon

Quick reference for the UI, CAD, slicing, AI, and QA terms used in this test plan. Only terms I actually use are listed here — if I drop a new one, ask anytime and I'll add it.

UI & web layout

Term	Plain-English meaning
Hero	The large intro block at the top of a marketing page (headline + image + CTA buttons). What you'd call the 'homepage banner'.
Header	The persistent top bar with the logo, nav links, and sign-in button. Stays put while the page scrolls.
Footer	The persistent bottom bar with copyright, secondary links, and contact info.
Banner	A wide horizontal info strip (often colored) used for announcements; dismissible with an X.
Toast / snackbar	A small auto-fading notification that pops into a corner (usually bottom-right) — 'success', 'error', 'tip-of-the-day' messages.
Modal / dialog	A popup window that takes focus and dims the page behind it; closes on X / ESC / outside-click.
Popover / tooltip	A small floating info bubble attached to a button or icon — not modal, dismisses on click-away.
Carousel / strip	A horizontally-scrolling row of items, usually with prev/next arrows. ForgeSlicer's Beginner Starters is NOT a carousel — it's a grid.
Grid	Items laid out in rows × columns. 'Responsive grid' means the column count changes with screen size.
Tab / tab bar / tab panel	A row of clickable tabs that swap the content area below. Active tab is the highlighted one; tab panel is the swap-able body.
CTA (Call-to-Action)	The primary clickable button you want the user to press — e.g. 'Start Designing Free'.
data-testid	An HTML attribute we add to elements specifically so automated tests can find them reliably (independent of styling changes).

3D editor (workspace)

Term	Plain-English meaning
Workspace	Our in-browser CAD editor at /workspace — where you build, edit, and slice models.
Viewport / canvas	The 3D view in the middle of the workspace where the model is shown.
Build plate	The orange rectangle on the floor of the viewport — represents your printer's bed; objects must fit inside it.
Primitive	A basic built-in shape (cube, sphere, cylinder, cone, torus, triangle) that you drop in and edit.
Gizmo	The colored arrows / handles attached to a selected object that you drag to move (translate), rotate, or scale it.
Face handle	The small colored square in the middle of each face of a selected primitive — drag it to resize that face directly, TinkerCAD-style.
Inspector / right panel	The right-hand sidebar showing the selected object's name + numeric dimensions / position / rotation / scale.
Outliner / scene tree	The left-hand list of every object in the scene with visibility, lock, duplicate, delete controls.
Transform	A position + rotation + scale combined — the 'where and how big' of an object.

Term	Plain-English meaning
Snapping	Locking transforms to a grid step (e.g. 1 mm position, 15° rotation) so you don't end up with 12.473 mm.
Modifier	An object marked 'positive' (adds to the final shape) or 'negative' (cuts away from the final shape) before booleans run.
Boolean operation	Union (merge), Subtract (cut), or Intersect (keep overlap) — the three ways to combine two meshes.
Mesh	A 3D shape made of triangles. STL files are pure meshes.
Manifold / watertight	A mesh with no holes, no self-intersections, and a clear inside/outside — required for reliable slicing.
Normal	A direction-vector that says 'this triangle is facing outward'. Inverted normals cause slicers to confuse inside vs outside.
Bounding box (bbox)	The smallest axis-aligned rectangular box that contains an object — used for fit checks against the build plate.

File formats & slicing

Term	Plain-English meaning
STL	Plain mesh file — just triangles. No colors, no units, no print settings. Most common 3D-printing format.
OBJ	Mesh file that can also carry material/group info. Older format from the 3D-art world.
3MF	Modern multi-object mesh container (zip-based). Preserves transforms, modifiers, units, and slicer settings — the format we hand off to OrcaSlicer.
SVG	2D vector format. We can extrude it into a 3D solid by giving it a depth.
G-code	The plain-text file your printer actually reads — line-by-line instructions: move here, extrude this much, etc. Output of a slicer.
Slicer	Software that converts a 3D mesh into G-code by cutting it into layers and planning toolpaths. ForgeSlicer does this in-browser, server-side, or hands off to your desktop slicer.
Slicer handoff	Exporting a 3MF and opening it in OrcaSlicer / Bambu Studio / PrusaSlicer to finish slicing there instead of in-browser.
FDM	Fused Deposition Modeling — the most common consumer 3D-printing process; melts plastic filament and extrudes it layer by layer.

AI, voice & reverse engineering

Term	Plain-English meaning
Meshy / Meshy.ai	Third-party AI service we use to turn a text prompt or image into a 3D mesh. Not built by us — we just integrate with their API.
Whisper	OpenAI's speech-to-text model. We use it to transcribe what you say into the mic.
Intent / JSON intent	After Whisper transcribes your voice, GPT-5.2 converts the sentence into a structured JSON action like '{op:"add",shape:"cube",size:20}' that the workspace executes.
RANSAC	RANdom SAmple Consensus — a statistical algorithm that looks at a messy imported mesh and detects clean shapes (boxes, cylinders, spheres) hiding inside it. The basis for our 'Reverse Engineer' feature.

Term	Plain-English meaning
Primitive fitting	The second half of RANSAC — once shapes are detected, we estimate their exact dimensions and transforms so we can swap the mesh for editable primitives.

QA & release process

Term	Plain-English meaning
P0 / P1 / P2	Priority levels. P0 = run every release (must pass to ship), P1 = run weekly (regression coverage), P2 = run when the related area changes.
Smoke test	A 30-second sanity check that the app boots and the headline features still work — not a full test pass.
Regression	A bug that re-appears in something that used to work. Test plans exist mainly to catch these.
iter-N	Our internal iteration tag (e.g. 'iter-108'). Shown next to the logo on the landing page and bumped each release.
Sandbox sender	Resend's onboarding@resend.dev address — only allowed to email the account owner until a custom domain is verified.

Sign-off

When all P0 cases pass and no P1 regressions are open, this build is ready to ship. Record the tester, date, browser, and any deviations below before archiving.

Tester	
Date	
Browser / OS	
Build / commit	
Notes	